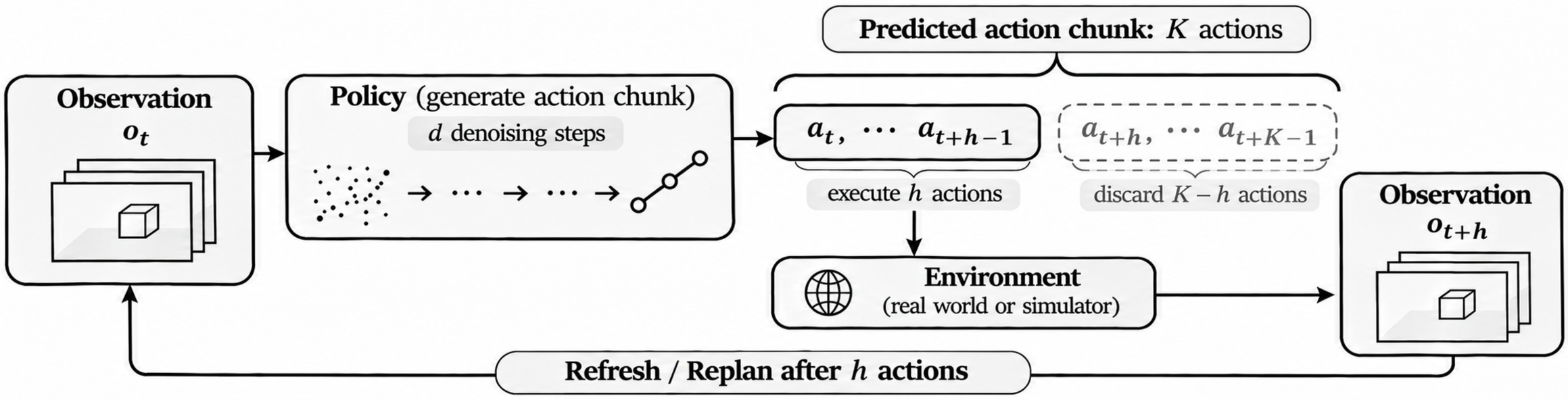
Refinement vs. Refresh

Compute Allocation for Closed-Loop Robot Policies

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Chunked Closed-Loop Policy



Parameters | K : number of predicted actions per chunk (chunk length) | h : executed actions per replan ($h \le K$) | d : denoising steps (refinement)

Synchronous closed-loop execution. Each policy call predicts K actions; only the first h are executed before obtaining a new observation and replanning.

Discovery
480 attempts · 12 settings
4 Spatial tasks

→ **Freeze**
3 denoising comparisons
before confirmation

→ **Confirmation**
1,120 attempts · 7 settings
8 tasks excluded from selection

Prefix contrasts: later-amended descriptive analyses

Setup (held-out)

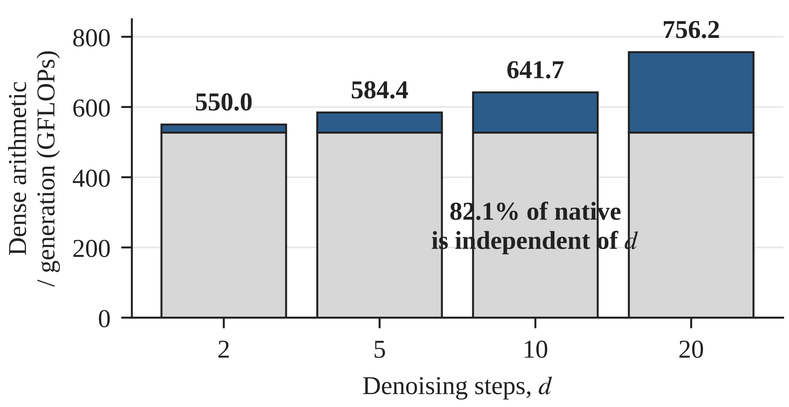
SmolVLA (frozen) $K = 50,\ d \in \{2, 5\}$

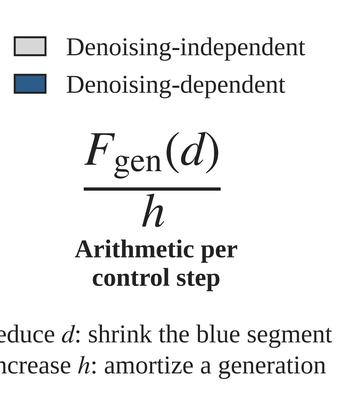
LIBERO (simulation) $h \in \{2, 5\}$ 8 tasks, 1,120 attempts

?

Under a fixed inference budget, should we allocate compute to additional denoising steps (**refinement**) or to more frequent observation-conditioned replanning (**refresh**)?

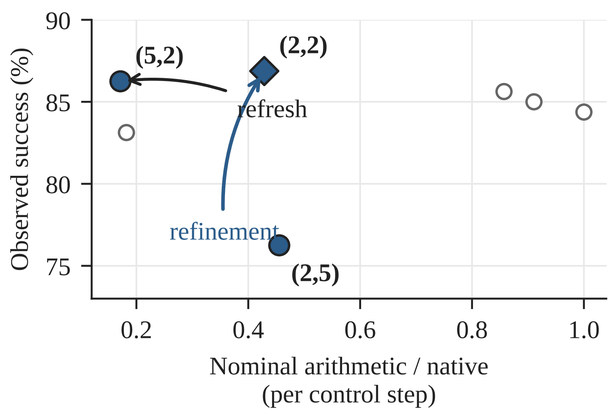
Why the Trade-Off Exists

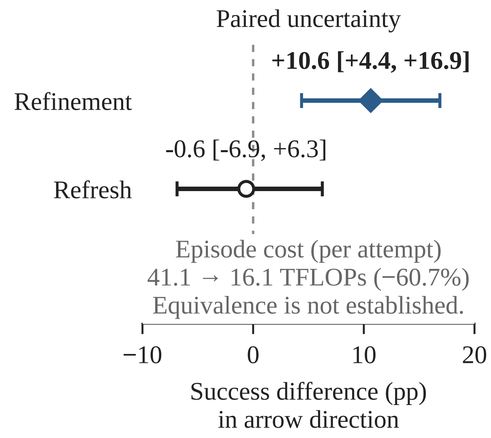




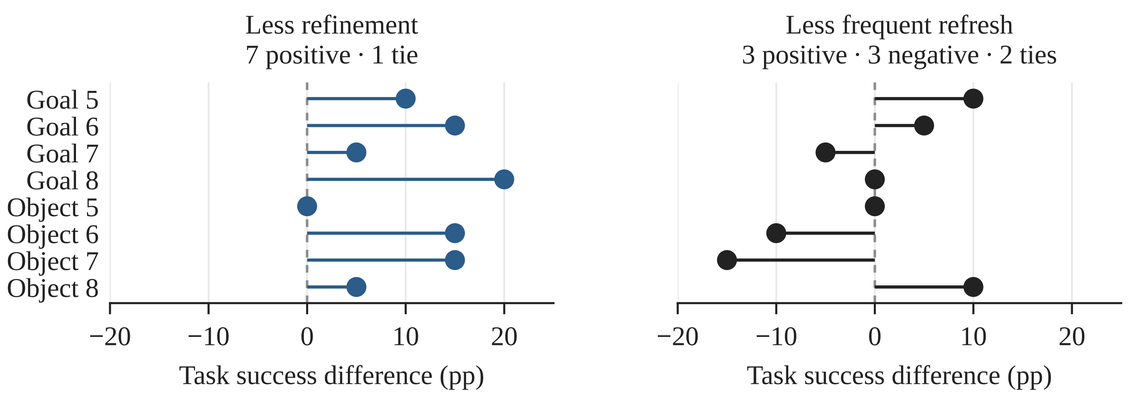
Reduce d: shrink the blue segment
Increase h: amortize a generation

Held-Out Success and Compute





What the Aggregate Results Hide



Primary refinement pairs: 24 wins / 7 losses. Task rows are descriptive; no task-level intervals are inferred.

General lesson

Use a within-model protocol: define refinement and refresh, measure complete generation cost, freeze the primary comparisons, then evaluate held-out success, expenditure, and task variation together.

Limits

One frozen checkpoint in synchronous simulation. Arithmetic is not deployment latency, and similar observed success does not establish equivalence.